

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Information Security	Course Code:	CS3002
	Program:	BCS	Semester:	Fall 2026
	Sections:	7H,7K	Total Marks:	10
	Due Date:	13/9/2026	Weight	~1.5%
	Exam Type:	Assignment 1	Page(s):	1

Cryptanalysis of Homophonic Cipher

You are provided with a file containing ciphertext that was generated by a homophonic cipher.

All homophones are made up of two characters: a lower case letter and a digit, like 'p9'. Tokens in ciphertext are separated via pipe | character. There are also some other non-homophonic tokens in ciphertext like spaces, punctuation, digits. These were transferred from plaintext unencrypted.

Tasks

- Write code to read the ciphertext file and do a frequency count of all homophones. Show the frequency counts as a bar/stem chart.
- Write code to calculate bigram frequencies and plot top 15 of them. If there is a non-homophonic character between two tokens, then don't consider that token pair as a bigram. For example, consider ciphertext y5|18|v6| |e8|r6|. Only valid bigrams are y5-18, 18-v6 and e8-r6.
- Similarly, calculate trigram frequency counts and plot top 10 of them.
- Make a guess of cipher key using bigram and trigram frequency information.

Deliverables

1. Program in C/C++ or Python. Submit a single source file for first three tasks.
2. Document containing key guess with rationale.